CLASS 9 · CBSE · GANITA MANJARI PART 1 · SHORT NOTES

Chapter 7: The Mathematics of Maybe: Introduction to Probability

Topic-wise short notes with solved examples

Probability measures how likely an event is, on a scale from 0 to 1. This chapter covers experimental and theoretical probability, sample spaces, events and tree diagrams.

1 Randomness and the probability scale

- A **random experiment** is an action that can be repeated, whose possible results are known but whose actual result cannot be predicted. Tossing a coin and rolling a die are examples.
- Each performance of the experiment is a **trial**, and each possible result is an **outcome**.
- **Probability** is a number from 0 to 1 that measures how likely an event is. For any event E, $0 \leq P(E) \leq 1$.

Probability	Meaning	Example
0	Impossible	Getting 7 on an ordinary die
Between 0 and $\frac{1}{2}$	Less likely	Getting 3 on a die
$\frac{1}{2}$	Even chance	Getting heads on a coin
Between $\frac{1}{2}$ and 1	More likely	Getting a number greater than 1 on a die
1	Certain	Picking a red sweet from a bag of only red sweets

- A probability can be written as a fraction, a decimal or a percentage: $\frac{3}{4} = 0.75 = 75\%$.
- A coin or die is **fair** (unbiased) when no outcome is favoured over another.
- A guess based on personal judgement ("it looks like rain") is a **subjective** probability. This chapter is about measuring probability **objectively**.

2 Experimental probability

- Experimental probability is found from actual trials or from recorded data.

$$\text{Experimental probability} = \frac{\text{Number of times the event occurred}}{\text{Total number of trials}}$$

- This fraction is also called the **relative frequency** of the event.
- Different sets of trials can give different values.

Example 2.1 A spinner is spun 80 times and stops on red 28 times. Find the experimental probability of red.

SOLUTION

$\frac{28}{80} = \frac{7}{20}$

0.35 **or 35%**

3 Theoretical probability

- Theoretical probability is found by reasoning, when all the outcomes are **equally likely**. No experiment is needed.

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Number of possible outcomes}}$$

- The event "E does not happen" has probability $P(\text{not } E) = 1 - P(E)$.

Example 3.1 A fair die is rolled once. Find the probability of getting (a) a prime number (b) a number that is not prime.

SOLUTION

The primes on a die are 2, 3 and 5: 3 favourable outcomes out of 6.

$P(\text{not prime}) = 1 - \frac{1}{2}$

(a) $\frac{1}{2}$ (b) $\frac{1}{2}$

Example 3.2 A letter is picked at random from the word MATHEMATICS. Find the probability that it is (a) M (b) a vowel.

SOLUTION

The word has 11 letters. M occurs twice. The vowels are A, E, A, I: four letters.

(a) $\frac{2}{11}$ (b) $\frac{4}{11}$

Example 3.3 A point is chosen at random inside a square of side 4 m. A circle of radius 1 m is drawn inside the square. Find the probability that the point lies inside the circle. Take $\pi = 3.14$.

SOLUTION

When a point is chosen at random in a region, the probability is the ratio of the areas.

$\frac{\pi \times 1^2}{4 \times 4} = \frac{3.14}{16}$

about 0.196

4 Probability from data and samples

- When data has been collected, the relative frequency is used as an estimate of the probability.
- Data is usually collected from a **sample**, a part of the whole **population**. The estimate from the sample is then applied to the population.
- A larger sample that represents all parts of the population gives a more reliable estimate.

Example 4.1 In a sample of 60 students, 18 walk to school. Estimate the probability that a student walks to school, and the number who walk in a school of 900 students.

SOLUTION

Probability $\approx \frac{18}{60} = 0.3$

Expected number $= 0.3 \times 900$

0.3; about 270 students

5 Experimental and theoretical probability compared

- For a small number of trials, the experimental probability can be quite different from the theoretical probability.
- **Law of Large Numbers**: as the number of trials grows, the experimental probability gets closer to the theoretical probability.
- **Gambler's fallacy**: believing that after several heads in a row, tails is "due". A fair coin has no memory. Each toss is independent, and the probability of tails stays $\frac{1}{2}$.

Example 5.1 A die is rolled 12 times and shows 3 on three of them. Compare the experimental and theoretical probabilities of getting 3. What do you expect for 6000 rolls?

SOLUTION

Experimental: $\frac{3}{12} = 0.25$. Theoretical: $\frac{1}{6} \approx 0.167$.

They differ because 12 trials are few. For 6000 rolls the experimental value should be close to $\frac{1}{6}$, that is about 1000 threes.

6 Sample space and events

- The **sample space** S is the list of **all** possible outcomes, each written once. The number of outcomes is written $n(S)$.
- An **event** is any outcome or group of outcomes. It is a subset of the sample space.
- The sample space must suit the question. For "will it rain?", $S = \{\text{Rain}, \text{No rain}\}$ is enough. To study how much it rains, more outcomes are needed.

Experiment	Sample space	$n(S)$
Tossing a coin	$\{H, T\}$	2
Rolling a die	$\{1, 2, 3, 4, 5, 6\}$	6
Tossing two coins	$\{HH, HT, TH, TT\}$	4
A coin and a die together	$\{H1, H2, \ldots, H6, T1, T2, \ldots, T6\}$	12

Example 6.1 A die is rolled. Write the event "a multiple of 3" as a set and find its probability.

SOLUTION

$E = \{3, 6\}$ and $n(S) = 6$

$P(E) = \frac{2}{6} = \frac{1}{3}$

Example 6.2 A stall offers 3 snacks (samosa, pakora, bhaji) and 2 drinks (chai, lassi). How many snack-and-drink choices are there? Find the probability that a choice made at random includes lassi.

SOLUTION

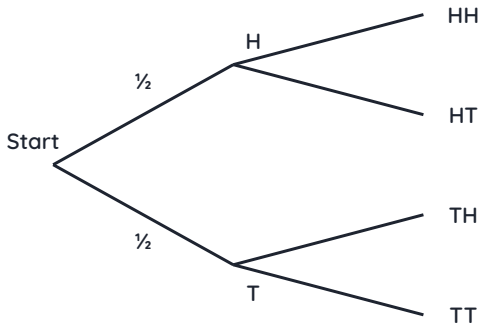
Each of the 3 snacks can go with each of the 2 drinks: $3 \times 2 = 6$ choices.

Three of the six choices include lassi.

6 choices; $\frac{3}{6} = \frac{1}{2}$

7 Tree diagrams

- A **tree diagram** lists the outcomes of an experiment done in steps. Each branch is one possible result of a step, and each path from the start to an end is one outcome.



Two tosses of a coin give four equally likely outcomes

- To find the probability of one path, **multiply** the probabilities along its branches. To combine several paths, **add** their probabilities.
- **With replacement**: the object is put back, so the probabilities are the same at each step. **Without replacement**: the numbers change after the first draw.

Example 7.1 A coin is tossed twice. Find the probability of (a) two heads (b) one head and one tail.

SOLUTION

$S = \{HH, HT, TH, TT\}$

(a) $\frac{1}{4}$ (b) $\frac{2}{4} = \frac{1}{2}$

Example 7.2 Three coins are tossed. Find the probability of (a) exactly two heads (b) at least one head.

SOLUTION

$S = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}$, so $n(S) = 8$.

Exactly two heads: HHT, HTH, THH. At least one head: every outcome except TTT.

(a) $\frac{3}{8}$ (b) $\frac{7}{8}$

Example 7.3 A bag has 2 red and 3 blue balls. A ball is drawn, put back, and a ball is drawn again. Find the probability that the two balls have the same colour.

SOLUTION

$P(\text{red, red}) = \frac{2}{5} \times \frac{2}{5} = \frac{4}{25}$ and $P(\text{blue, blue}) = \frac{3}{5} \times \frac{3}{5} = \frac{9}{25}$

$\frac{4}{25} + \frac{9}{25} = \frac{13}{25}$

Example 7.4 A bag has 3 red and 2 blue balls. Two balls are drawn one after the other without replacement. Find the probability of (a) red then blue (b) two blue balls.

SOLUTION

(a) $\frac{3}{5} \times \frac{2}{4} = \frac{6}{20}$. After a red ball is removed, 4 balls remain and 2 are blue.

(b) $\frac{2}{5} \times \frac{1}{4} = \frac{2}{20}$. After a blue ball is removed, only 1 blue ball remains.

(a) $\frac{3}{10}$ (b) $\frac{1}{10}$

8 Common mistakes

- Giving a probability greater than 1 or less than 0. Every probability lies from 0 to 1.
- Using the formula for theoretical probability when the outcomes are not equally likely (for example, "the car starts or does not start").
- Treating HT and TH as one outcome. They are different outcomes.
- Counting repeated letters once: in MATHEMATICS there are 11 letters, not 8.
- Forgetting that the total changes when a ball is not put back.
- The gambler's fallacy: past results do not change the probability of the next independent trial.